Lab 3: Architectural Style Selection & Justification

Incident Escalation & On-Call Rotation Engine

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Architectural Choice

The system is organized around events such as Alert Received, Engineer Assigned, Incident Acknowledged, and Escalation Timeout. An Alert Ingestion Component publishes events to an Event Bus, while the On-Call Rotation, Escalation, Notification, and Incident Tracking components react independently. This keeps alert processing asynchronous and allows each component to handle its specific responsibility.

Two Reasons for This Choice

1. **Natural fit for incident workflows:** The system must react to incoming alerts and trigger further actions when an engineer does not acknowledge an incident within the required time. Event-driven architecture models this sequence naturally through events and independent handlers.
2. **Flexible escalation and notifications:** Alert ingestion, rotation lookup, escalation, and notification can operate independently. New escalation levels or notification channels such as SMS, email, and webhooks can be added with minimal impact on the other components.

Security Advantage

Components can validate and authorize events before processing sensitive incident information. Access to rotation schedules and incident records can be controlled centrally, while only trusted components are allowed to publish or consume operational events. This helps protect engineer and incident data.

Performance Benefit

Asynchronous event processing allows alert ingestion to remain fast even when notifications or escalations take longer to complete. Multiple event consumers can process work independently, supporting quick alert dispatch and better handling of simultaneous incidents during peak load.

DIAGRAM

